U-tad
CENTRO UNIVERSITARIO
DE TECNOLOGÍA Y ARTE DIGITAL

BACHELOR FINAL PROJECT

PRELIMINARY PROJECT

STUDENT'S NAME

Senén Marqués

PROVISIONAL TITLE OF THE BFP

From Plate to Macros: Estimating Calories and Macronutrients from Food Images

TUTOR'S NAME

Javier Redondo Antón

PROJECT DESCRIPTION AND OBJECTIVES

With this project, we aim to build a pipeline that (1) detects and segments foods on a plate, (2) classifies each item, (3) estimates mass via volume (either monocular depth or scale-from-plate), and (4) converts mass to calories/macros using an open nutrition API.
Our contribution will be:
a) a calibrated, reproducible portion-estimation module with uncertainty,
b) a robustness study across lighting/occlusion, and
c) an ablation comparing two volume strategies (single-image scale vs. monocular depth).

Prior work shows classification datasets are mature (Food-101, UEC-Food) while portion estimation is the hard part; recent datasets (Nutrition5k, Food Portion Benchmark, MM-Food-100K) and methods (reference objects, monocular depth) show the potential of this project.

SIGNATURE OF THE TUTOR